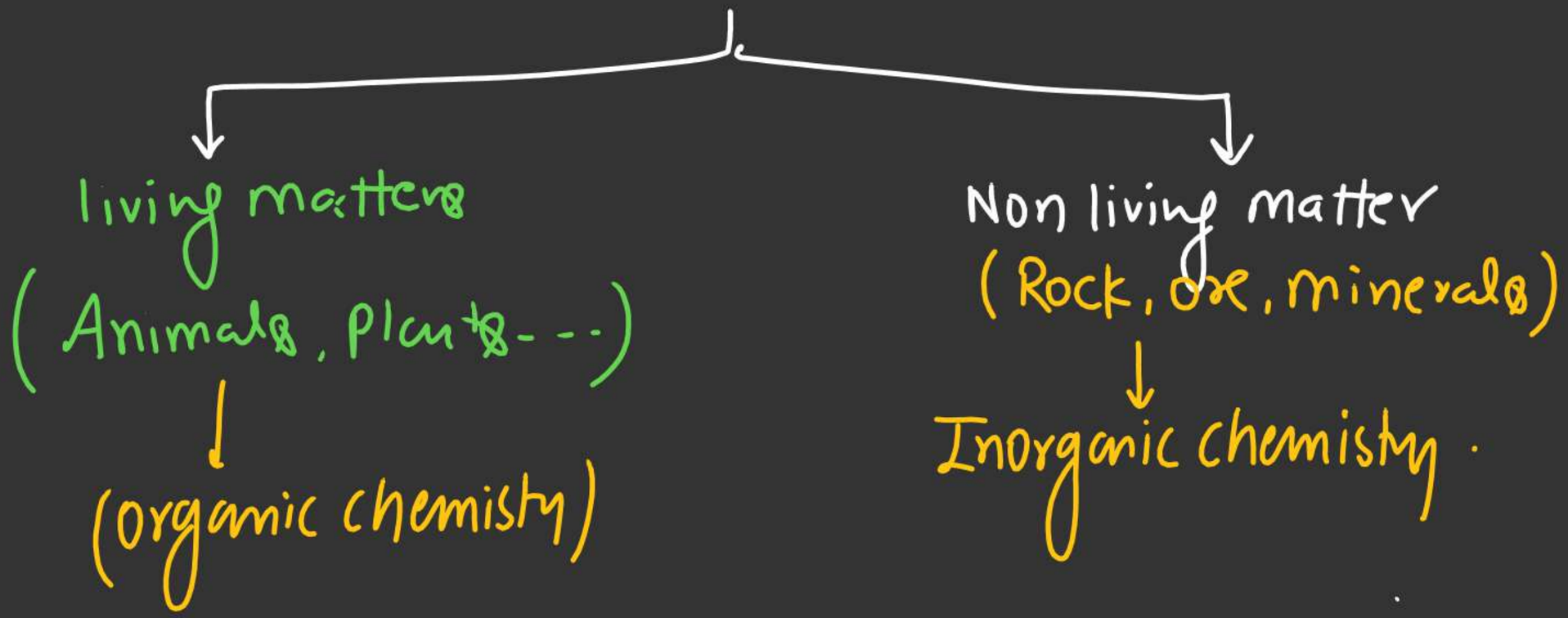


BASIC ORGANIC CHEMISTRY

SKM SIR (SURENDRA MISHRA SIR)

Organic Chemistry

John Jacob Berzilines



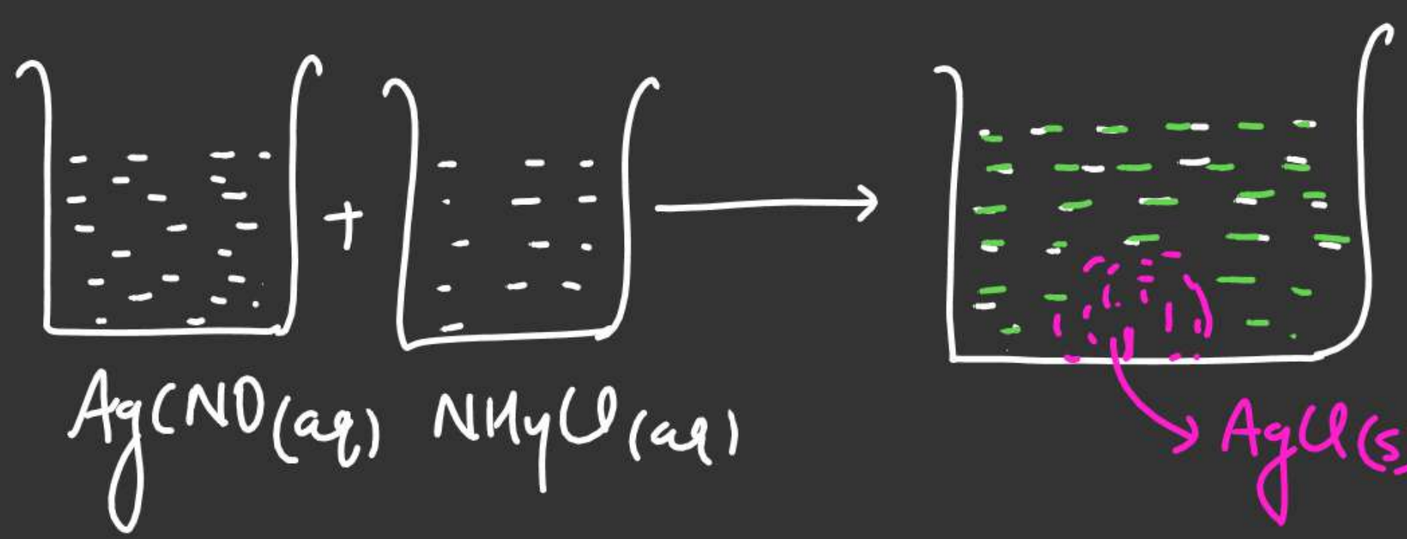
Acc to Berzilens: Organic Compounds are present in living matters like plants, animals..... & they are formed under influence of vital force. (vital force theory)

$\Rightarrow$ Organic Compounds can't be prepared in laboratory.

Wholar:

Silver Cynate + Ammonium chloride $\longrightarrow$ Ammonium Cynate + Silver chloride

$Ag^{+}CNO^{-}$ (Inorganic compond) + $NH_4^{+}Cl^{-}$ (Inorganic compond) $\longrightarrow$ NH_4CNO (Inorganic compond) + $AgCl$ (Inorganic compond)



Expected $NH_4^+CNO^-$

Observed $NH_2-\underset{\underset{O}{\|}}{C}-NH_2$ (Urea)

$\Rightarrow$ (Organic Compound)

It Concludes

(i) Organic Compounds can be prepared in laboratory

(ii) Violet vital force theory.

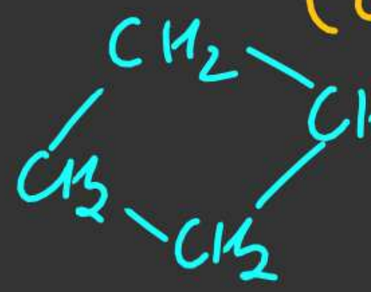
(iii) Phenomenon of isomerism

NH_4CNO $\quad$ $NH_2-\underset{\underset{O}{\|}}{C}-NH_2$

M.F. N_2H_4CO $\quad$ N_2H_4CO

(*) Study of Carbon & its compounds are known as Organic chemistry (Except Carbon Oxides (CO, CO_2, C_2O_3 ----))

Ex:

CH_4, CH_3-Cl, CH_3-CH_2-OH, 

(#) Representation method:

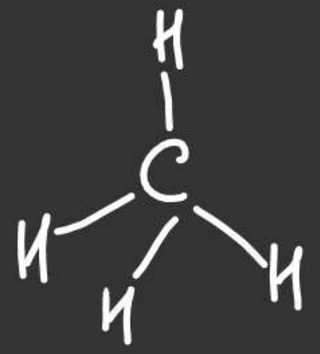
Note: (i) $-H, -D, -T, -F, -Cl, -Br, -I$ are monovalent (एक बांड बनाते हैं)

(ii) O, S usually are Bivalent (दो बांड बनाते हैं)

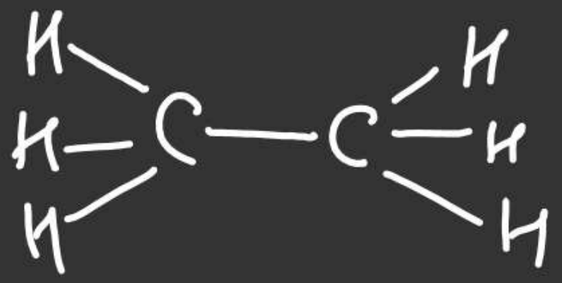
(iii) N Trivalent (तीन बांड बनाते हैं)

(iv) Carbon Tetravalent (चार बांड बनाता है)

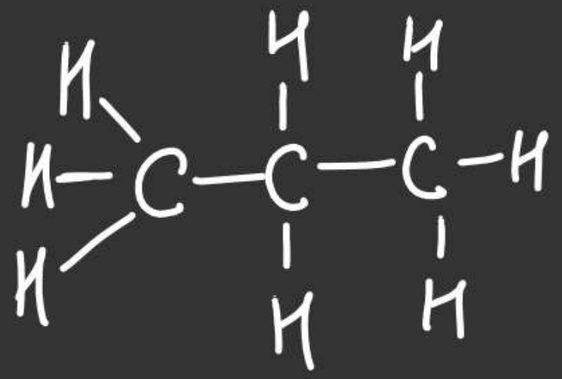
Ex: (1)



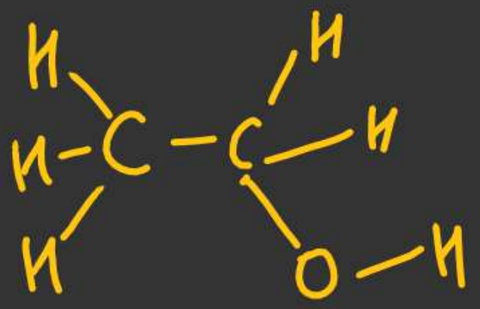
(2)



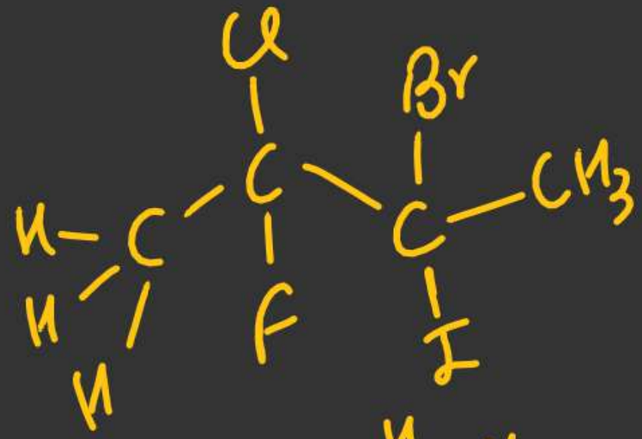
(3)



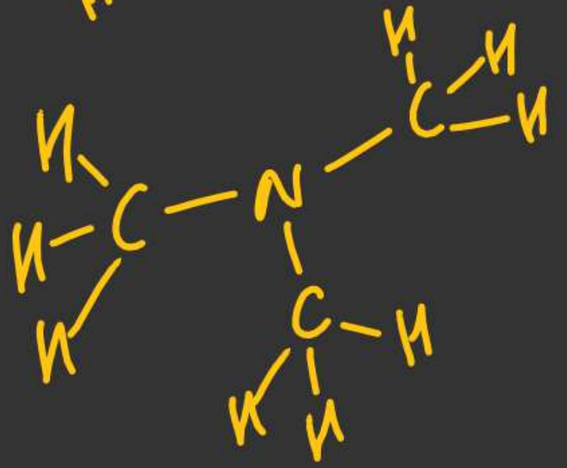
(4)

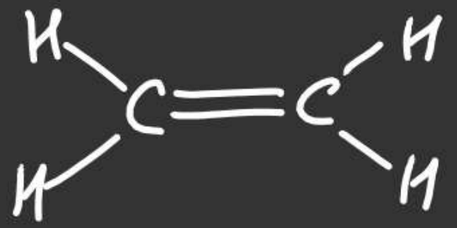


(5)



(6)



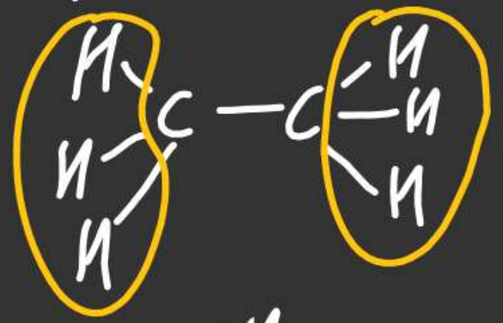
(7)  (2C & 4 H)

(8) H—C≡C—H (2C & 2H)

(1) Bond line Formula:

(a) Always draw Carbon Carbon Bond By a line.
(b) Never show C–H Bond
(C) Draw all Bonds Other than C-H

Note: Each Terminal & Corner wd represent Carbon atom

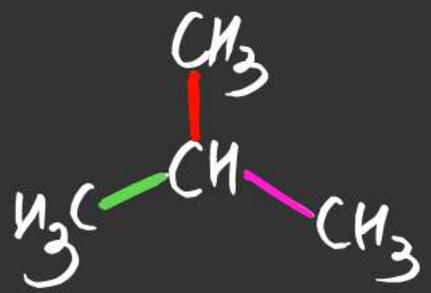
Ex-1:  or $H_3C—CH_3$ or —

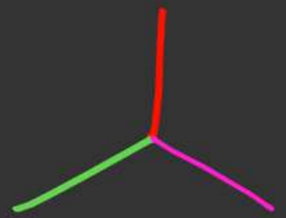
Ex-2: $H_3C-CH_2-CH_3$ (or) 

Ex-3:

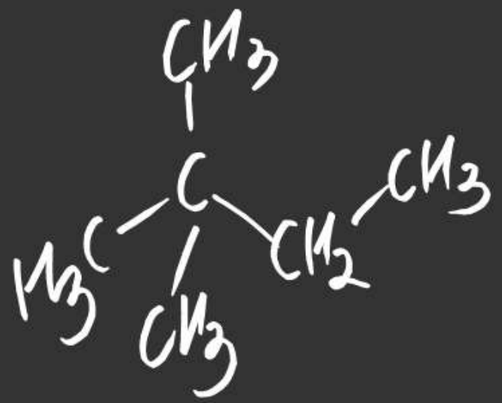


Ex-4:

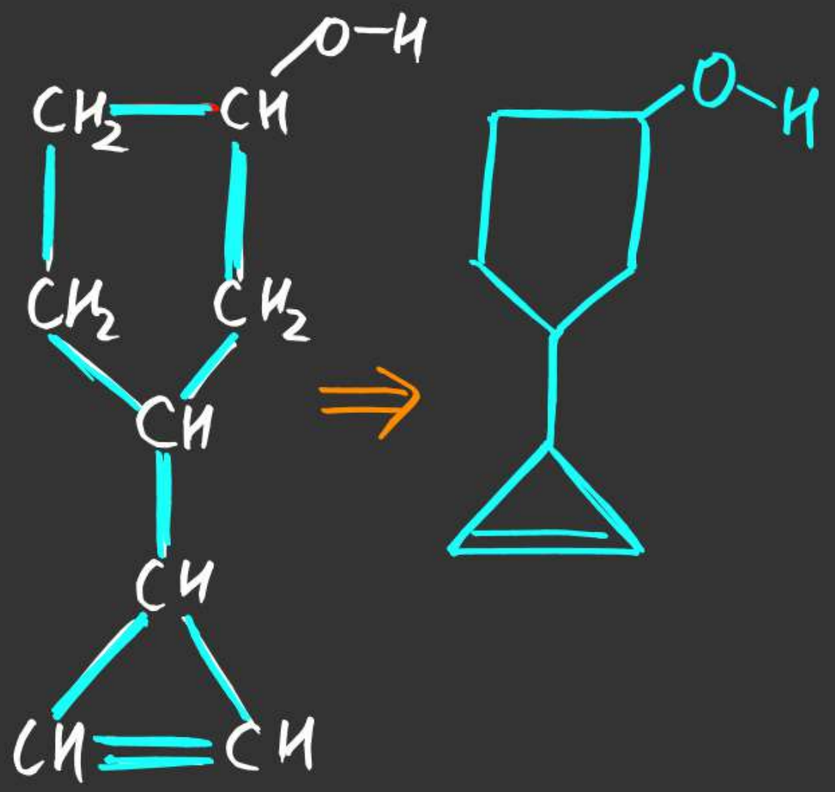




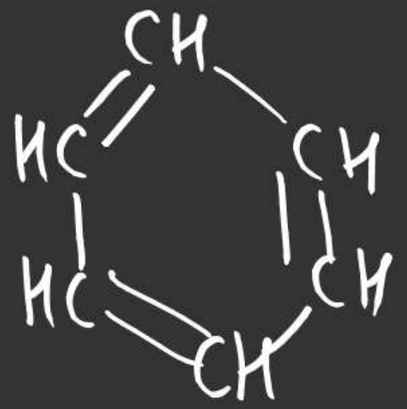
Ex-5:



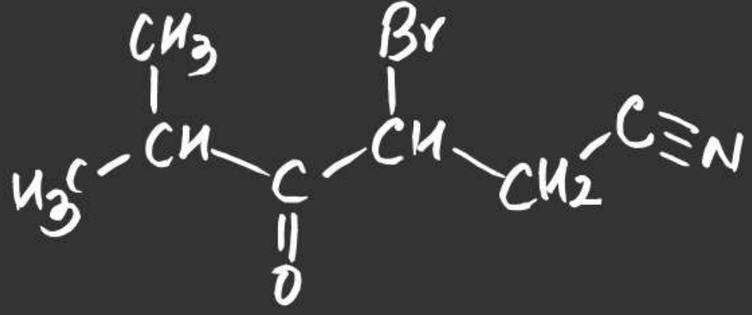
Ex-6:



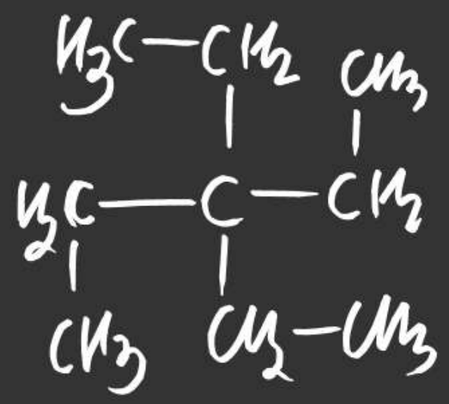
Ex-7:



Ex-8:

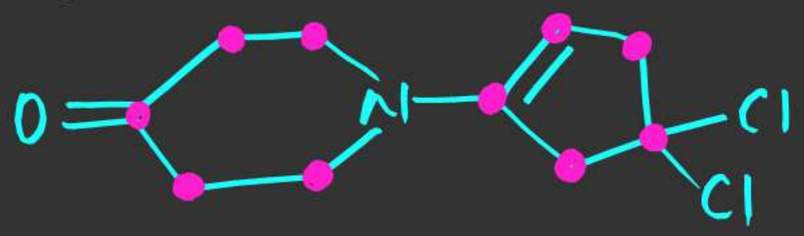


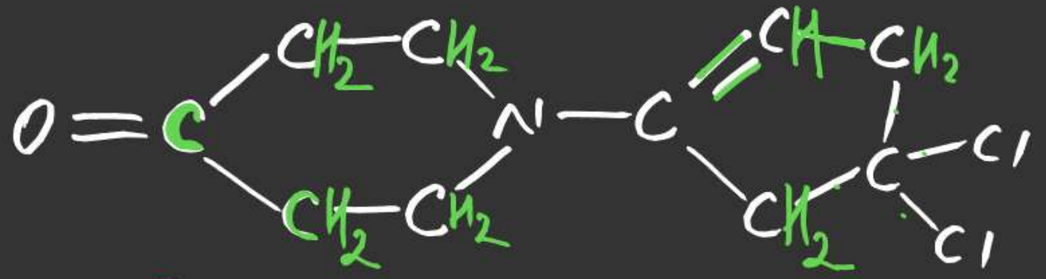
Ex-9:



Find Molecular Formula of following.

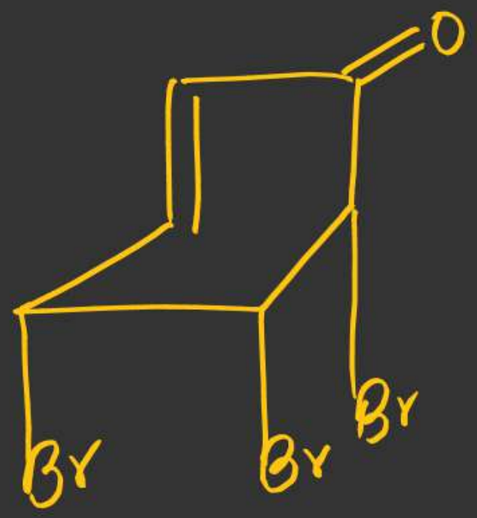
(10)





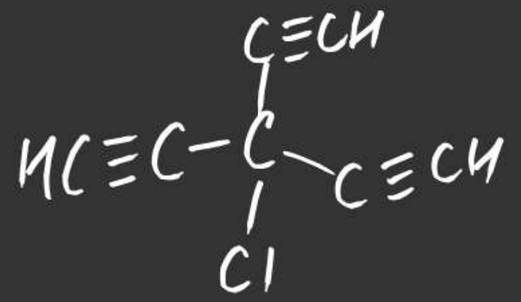
$(C_{10}H_{13}Cl_2NO)$

(11)

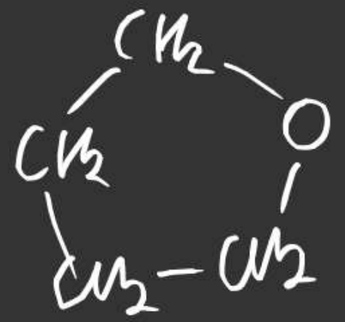


(12) SKM

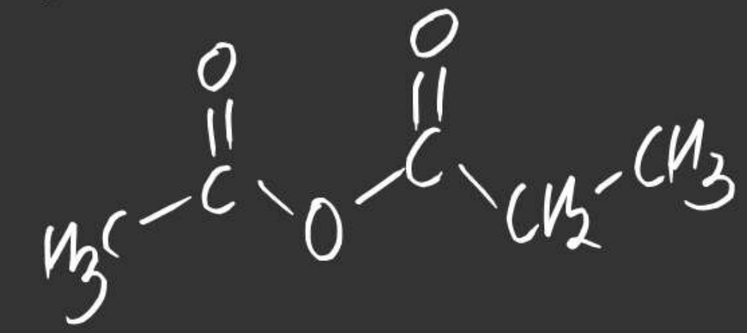
(13)



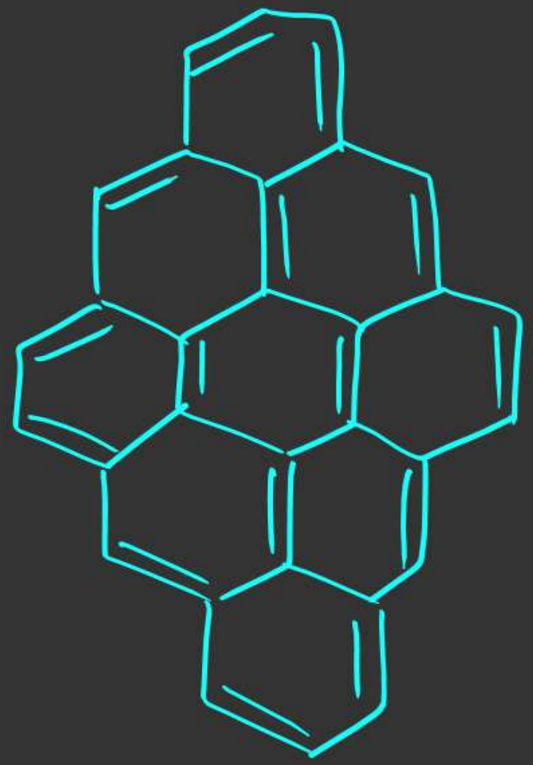
(14)



(15)

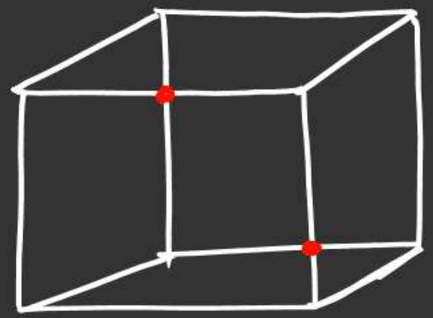


(16)



[Dibenzcorrene]

(17)



(Cube)

(18)

